

Science for Technology

Pythagorean Relationship (පයිතගරස් සම්බන්ධතාව)

Short Notes 03

Tech Hub

01. Right-Angled Triangle Fundamentals

[Diagram: Triangle ABC with Right Angle at B and Angle C]

A triangle containing one internal angle equal to 90 degrees (right angle).

- **Opposite Side (සම්මුඛ පාදය):** AB (opposite to reference angle C)
- **Adjacent Side (බද්ධ පාදය):** BC (adjacent to reference angle C)
- **Hypotenuse (කර්ණ පාදය):** AC (longest side, opposite 90°)

03. Classification of Triangles by Side Lengths

For any triangle with side lengths a, b, c where b is the longest side ($b > a, c$):

Geometric Condition	Triangle Type (Sinhala)	Triangle Type (English)
$b^2 < a^2 + c^2$	සුළුකෝණී ත්‍රිකෝණය	Acute-Angled Triangle
$b^2 = a^2 + c^2$	සෘජුකෝණී ත්‍රිකෝණය	Right-Angled Triangle
$b^2 > a^2 + c^2$	මහා කෝණී ත්‍රිකෝණය	Obtuse-Angled Triangle

04. Pythagorean Triples

A set of three positive integer values that satisfy the Pythagorean theorem.

Verification Example (3, 4, 5):
 $3^2 + 4^2 = 9 + 16 = 25 = 5^2$

- Any integer multiple of a fundamental triple is also a valid triple (derived triples).
- Enables quick right-angle verification in construction.

06. Standard Pythagorean Triples Reference

Column 1	Column 2	Column 3
(3, 4, 5)	(5, 12, 13)	(6, 8, 10)
(8, 15, 17)	(9, 40, 41)	(11, 60, 61)
(12, 35, 37)	(13, 84, 85)	(16, 63, 65)
(20, 21, 29)	(28, 45, 53)	—

02. Pythagoras' Theorem

In any right-angled triangle, the area of the square on the hypotenuse is equal to the sum of the areas of the squares on the other two sides.

Theorem Statement:

$$AB^2 + BC^2 = AC^2 \quad | \quad a^2 + b^2 = c^2$$

- Applies **exclusively** to right-angled triangles.
- Allows calculating any missing side when two side lengths are known.

05. Triple Generation Formula

Generates integer triples using positive integers a and b where $a > b$:

Generative Set: $(a^2 - b^2, 2ab, a^2 + b^2)$

Example (for $a = 2, b = 1$):

- Side 1: $2^2 - 1^2 = 3$
- Side 2: $2 \times 2 \times 1 = 4$
- Hypotenuse: $2^2 + 1^2 = 5 \rightarrow$ Triple: (3, 4, 5)

07. Converse of Pythagoras' Theorem

If the side lengths of a given triangle yield a valid Pythagorean triple ($a^2 + b^2 = c^2$), then the triangle is guaranteed to be a right-angled triangle.

08. Technical & Field Applications

- **Land Surveying:** Measuring perpendicular angles in land mapping.
- **Foundation Layout:** Setting up string lines for 90° building corners.
- **Roof Framing:** Determining rafter lengths, heights, and pitch.
- **Masonry:** Checking wall verticality and perpendicular joints.